Grade: Check Plus

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Assignment #1

x	$f(x)$	$x\,f(x)$	$(x-50)^2 f(x)$
10	0.10	1	160
20	0.15	3	135
40	0.30	12	30
60	0.15	9	15
80	0.25	20	225
100	0.05	5	125
		50	690

a.) $P(X \geq 60) = f(60) + f(80) + f(100)$

✓ Qa

$= 0.45$ or 45%

b.) $\mu = \sum_{\text{all } x} x\,f(x) = 10(0.10) + 20(0.15) + 40(0.30) + 60(0.15) + 80(0.25) + 100(0.05)$

$\mu = \$50$ expected value

$\sigma^2 = \sum_{\text{all } x} (x-50)^2 f(x)$

✓ Qb

$= (10-50)^2(0.10) + (20-50)^2(0.15) + \ldots$

$\sigma^2 = 690$

$\sigma = \sqrt{690} = \$26.27$ standard Dev.

c.) The expected value does not necessarily mean it's a possible outcome. It's the weighted average, representing the center of the probability distribution.

✓ Qc

d.) $P(50 - 26.27 \leq X \leq 50 + 26.27)$

$P(23.73 \leq X \leq 76.27)$

$P(40 \leq X \leq 60) = 0.30 + 0.15$

✓ Qd

$= 0.45$ or 45%

e.) $P(50 - 2(26.27) \leq X \leq 50 + 2(26.27))$

✓ Qe

$P(-2.54 \leq X \leq 102.54)$

$P(10 \leq X \leq 100) = 100\%$